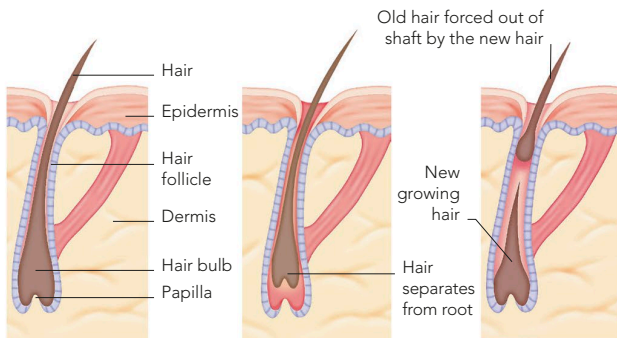


HAIR GROWTH

Hairs are rods of dead cells filled with keratin. The root, or bulb, is buried in a pit, the follicle. As extra cells add to the root, the hair lengthens. Different kinds of hairs grow at varying rates, with scalp hairs lengthening about 0.3mm ($\frac{1}{100}$ in) each day. Hair does not grow continuously. After three to four years, the follicle goes into a rest phase and the hair may fall out. Three to six months later, the follicle activates again and begins to produce a new hair.



ACTIVELY GROWING

New cells created at the root get pushed up so the hair gets longer.

INACTIVE PHASE

Activity in the follicle stops and the hair stops growing.

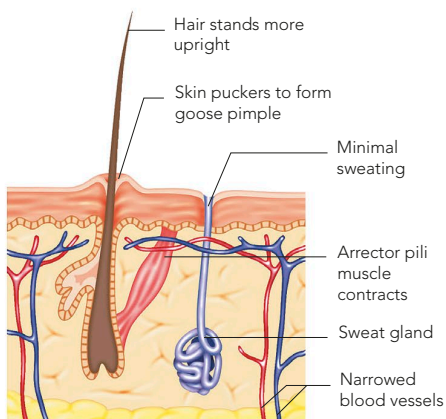
NEW GROWTH

The follicle root reactivates and starts to produce a new hair as the old one falls out.

TEMPERATURE REGULATION

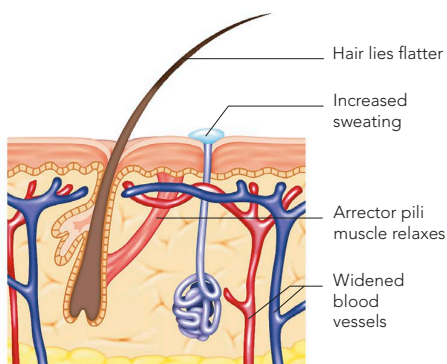
One of the skin's functions is to help maintain a constant body temperature. If the body is too hot, blood vessels in the dermis widen (vasodilate) to allow extra blood flow so more heat can be lost from the surface, and sweat glands produce more

sweat, which evaporates, drawing away body heat. If the body is cold, the skin's blood vessels narrow (vasoconstrict) to minimize heat loss, and sweating is reduced. Tiny hairs are pulled upright by the arrector pili muscles to trap an insulating layer of air.



FEELING COLD

Tiny body hairs are raised, creating goose pimples at their bases. The skin's blood vessels constrict, reducing blood flow, and sweat glands reduce their activity.



FEELING HOT

Tiny body hairs lie flatter and the goose pimples disappear. The blood vessels widen, increasing blood flow, and sweat glands increase sweat production.